

Alena Cheskidova
Email: t3531350@yahoo.com

To: Quantitative Economics

Cover letter

Dear Editors,

We are pleased to submit our article entitled «Increment Based Correlation» for your consideration.

This work proposes a new dependence measure, the Cheskidov Correlation (CIC or Cheskidov iCorrelation), which quantifies synchrony of directional increments in ordered data, with a particular focus on econometric time-series applications such as financial returns, volatility modeling, and macroeconomic indicators. The measure addresses a fundamental gap in econometric correlation analysis by providing a consistent way to capture true dependence in nonlinear, oscillatory, and other complex time-series settings—common in asset pricing, business cycles, and economic forecasting—where standard measures such as Pearson's correlation, Spearman's rho, and Kendall's tau often yield misleading or uninformative results due to their sensitivity to mean shifts or rank distortions. The main contributions are:

- Introduction of a new correlation coefficient based on normalized forward differences of ordered observations, directly applicable to differenced economic time series.
- A theoretical characterization of CIC within a Cauchy–Schwarz–type framework, ensuring clear probabilistic and geometric interpretation suitable for econometric theory.
- Analytical comparison with classical correlation coefficients, highlighting conditions under which CIC coincides with or diverges from Pearson's and rank-based measures in economic contexts.
- Simulation and empirical illustrations using financial and macroeconomic data, showing that CIC avoids spurious correlations that commonly arise in oscillatory or symmetric nonlinear time-series relevant to econometrics.

The results are original, have not appeared in any prior publication or conference proceedings, and the article is not under consideration elsewhere. All authors have read and approved the submission and confirm adherence to your journal Ethical Principles for Authors. The manuscript is not a revised or resubmitted version of a previously rejected submission.

If accepted, we will provide fully annotated code and data used to generate the article, examples and figures. All real or synthetic data used in the paper will be supplied as Supplementary Material in accordance with journal requirements. The Supplement is written in the same format as the article and contains complete technical details.

All co-authors are willing to serve as referees for future submissions.

We believe that the proposed methodology may be of interest to a broad statistical audience, particularly those working with nonlinear dependence, time-series analysis, signal analysis and measures of association. The paper could also be considered for the Discussion Paper track because it introduces a new dependence measure and addresses a longstanding conceptual issue in statistical methodology.

Thank you for your consideration.

Sincerely yours,

Alena Cheskidova

Ural Federal University

Email: alyona.cheskidova@gmail.com

Roman Cheskidov

Independent Researcher

Email: t3531350@yahoo.com

Anna Cheskidova

Independent Researcher

Email: anya2.cheskidova@gmail.com